

oct. 30, 18 18:36

AddMemberTest.java

Page 1/4

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package tests;

import opinion.ISocialNetwork;
import opinion.SocialNetwork;

import exceptions.BadEntryException;
import exceptions.MemberAlreadyExistsException;
import exceptions.NotTestReportException;

/**
 * Tests for the SocialNetwork.<i>addMember()</i> method.
 *
 * @author B. Prou, E. Cousin, GO
 * @version V2.0 - April 2018
 */
public class AddMemberTest {

    /**
     * Check that trying to add this new member (login, pwd, profile) raises a
     * BadEntry exception and does not change content of the
     * <i>ISocialNetwork</i>. If OK, the method just returns 0. If not OK,
     * displays an error message and returns 1.
     *
     * @param sn
     *      - the <i>ISocialNetwork</i>
     * @param login
     *      - new member's login
     * @param pwd
     *      - new member's password
     * @param profile
     *      - new member's profile
     * @param testId
     *      - the test ID that will prefix any error message displayed by
     *        this method
     * @param errorMessage
     *      - the error message that will be displayed if no exception is
     *        thrown when adding this member
     * @return 0 if the test is OK, 1 if not
     */
    private static int addMemberBadEntryTest(ISocialNetwork sn, String login,
        String pwd, String profile, String testId, String errorMessage) {

        int nbMembers = sn.nbMembers(); // Number of members when starting to
        // run this method

        try {
            sn.addMember(login, pwd, profile); // Try to add this member
            // Reaching this point means that no exception was thrown by
            // addMember()
            System.out.println("Err " + testId + ":" + errorMessage); // display
                                                                    // the
                                                                    // error
                                                                    // message

            return 1; // and return the "error" value
        } catch (BadEntryException e) { // BadEntry exception was thrown by
            // addMember() : this is a good start!
            // Let's now check if 'sn' was not
            // impacted
            if (sn.nbMembers() != nbMembers) { // The number of members has
                // changed : this is an error
                // case.

                System.out
                    .println("Err "
                        + testId
                        + " : BadEntry was thrown but the number of members was changed"); // Display
                                                                    // a
                                                                    // specific
                                                                    // error
                                                                    // message

                return 1; // return "error" value
            } else
                // The number of members hasn't changed, which is considered a
                // good indicator that 'sn' was not modified
                return 0; // return success value : everything seems OK, nothing
                // to display
        } catch (Exception e) { // An exception was thrown by addMember(), but
            // it was not the expected exception BadEntry
            System.out.println("Err " + testId + " : unexpected exception. "
                + e); // Display a specific error message
            e.printStackTrace(); // Display contextual info about what happened
            return 1; // return error value
        }
    }

    /**
     * Check that trying to add this new member (login, pwd, profile) raises an
     * AlreadyExists exception and does not change content of the
     * <i>ISocialNetwork</i>. If OK, the method just returns 0. If not OK,
     * displays an error message and returns 1.
     *
     * @param sn
     *      - the <i>ISocialNetwork</i>

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oct. 30, 18 18:36

AddMemberTest.java

Page 2/4

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* @param login
*         - new member's login
* @param pwd
*         - new member's password
* @param profile
*         - new member's profile
* @param testId
*         - the test ID that will prefix any error message displayed by
*           this method
* @param errorMessage
*         - the error message that will be displayed if no exception is
*           thrown when adding this member
* @return 0 if the test is OK, 1 if not
*/
private static int addMemberAlreadyExistsTest(ISocialNetwork sn,
String login, String pwd, String profile, String testId,
String errorMessage) {
    int nbMembers = sn.nbMembers(); // Number of members when starting to
    // process this method
    try {
        sn.addMember(login, pwd, profile); // Try to add this member
        // Reaching this point means that no exception was thrown by
        // addMember()
        System.out.println("Err " + testId + ":" + errorMessage); // display
                                                                    // the
                                                                    // error
                                                                    // message
        return 1; // and return the "error" value
    } catch (MemberAlreadyExistsException e) { // AlreadyExists exception was
        // thrown by addMember() :
        // this is a good start!
        // Let's now check if 'sn'
        // was not impacted
        if (sn.nbMembers() != nbMembers) {
            System.out
                .println("Err "
                    + testId
                    + ": MemberAlreadyExists was thrown, but the number of members was changed"); // Display
                                                                    // a
                                                                    // specific
                                                                    // error
                                                                    // message

            return 1; // and return the "error" value
        } else
            return 0; // return success value : everything is OK, nothing to
            // display
    } catch (Exception e) { // An exception was thrown by addMember(), but
        // it was not the expected exception
        // AlreadyExists
        System.out.println("Err " + testId + ": unexpected exception. "
            + e); // Display a specific error message
        e.printStackTrace(); // Display contextual info about what happened
        return 1; // return error value
    }
}

/**
* Check that this new member (login, pwd, profile) can be (and <i>is</i>)
* added to the <i>ISocialNetwork</i>.</br> If OK, the method just returns 0
* : the new member was added.</br> If not OK, an error message is displayed
* and 1 is returned ; the new member was not correctly added.
*
* @param sn
*         - the <i>ISocialNetwork</i>
* @param login
*         - new member's login
* @param pwd
*         - new member's password
* @param profile
*         - new member's profile
* @param testId
*         - the test ID that will prefix any error message displayed by
*           this method
* @return 0 if the test is OK, 1 if not
*/
private static int addMemberOKTest(ISocialNetwork sn, String login,
String pwd, String profile, String testId) {
    int nbMembers = sn.nbMembers(); // Number of members when starting to
    // process this method
    try {
        sn.addMember(login, pwd, profile); // Try to add this member
        // No exception was thrown. That's a good start !
        if (sn.nbMembers() != nbMembers + 1) { // But the number of members
            // hasn't changed
            // accordingly
            System.out.println("Err " + testId
                + ": the number of members (" + nbMembers
                + ") was not incremented"); // Error message displayed
            return 1; // return error code
        } else
            return 0; // return success code : everything is OK, nothing to
            // display
    } catch (Exception e) { // An exception was thrown by addMember() : this

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        // is an error case
        System.out
            .println("Err " + testId + ":unexpected exception " + e); // Error
                                                                    // message
                                                                    // displayed
        e.printStackTrace(); // Display contextual info about what happened
        return 1; // return error code
    }
}

/**
 * <i>addMember()</i> main test :
 * <ul>
 * <li>check if members can be added</li>
 * <li>check if incorrect parameters cause addMember() to throw BadEntry
 * exception</li>
 * <li>check if adding already registered members cause addMember() to throw
 * AlreadyExists exception</li>
 * </ul>
 *
 * @return a summary of the performed tests
 */
public static TestReport test(){

    ISocialNetwork sn = new SocialNetwork();

    int nbBooks = sn.nbBooks(); // number of books in 'sn' (should be 0
                                // here)
    int nbFilms = sn.nbFilms(); // number of films in 'sn' (should be 0
                                // here)

    int nbTests = 0; // total number of performed tests
    int nbErrors = 0; // total number of failed tests

    System.out.println("Testing addMember()");

    // <=> test nÂ°1

    // check if incorrect parameters cause addMember() to throw BadEntry
    // exception

    nbTests++;
    nbErrors += addMemberBadEntryTest(sn, null, "qsdgh", "", "1.1",
        "addMember() doesn't reject null logins");
    nbTests++;
    nbErrors += addMemberBadEntryTest(
        sn,
        " ",
        "qsdgh",
        "",
        "1.2",
        "addMember() doesn't reject logins that don't contain at least one character other than space");
    nbTests++;
    nbErrors += addMemberBadEntryTest(sn, "B", null, "", "1.3",
        "addMember() doesn't reject null passwords");
    nbTests++;
    nbErrors += addMemberBadEntryTest(
        sn,
        "B",
        " qwd ",
        "",
        "1.4",
        "addMember() doesn't reject passwords that don't contain at least 4 characters (not taking into account leading or trailing blanks)");
    nbTests++;
    nbErrors += addMemberBadEntryTest(sn, "BBBB", "bbbb", null, "1.5",
        "addMember() doesn't reject null profiles");

    // <=> test nÂ°2

    // populate 'sn' with 3 members

    nbTests++;
    nbErrors += addMemberOKTest(sn, "Paul", "paul", "lecteur impulsif",
        "2.1a");
    nbTests++;
    nbErrors += addMemberOKTest(sn, "Antoine", "antoine",
        "grand amoureux de la litt  rature", "2.1b");
    nbTests++;
    nbErrors += addMemberOKTest(sn, "Alice", "alice",
        "passionn  e de bande dessin  e", "2.1c");

    // try to add already registered members

    nbTests++;
    nbErrors += addMemberAlreadyExistsTest(sn, new String("Paul"),
        "abcdefghij", "", "2.2",
        "The login of the first member was accepted as login for a new member");
    nbTests++;
    nbErrors += addMemberAlreadyExistsTest(sn, new String("Alice"),
        "abcdefghij", "", "2.3",
        "The login of the last member was accepted as login for a new member");
    nbTests++;
    nbErrors += addMemberAlreadyExistsTest(

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        sn,
        new String("anToine"),
        "abcdefghij",
        "",
        "2.4",
        "An already registered login, but with different case, was accepted as login for a new member" );
nbTests++;
nbErrors += addMemberAlreadyExistsTest (
    sn,
    new String(" Antoine "),
    "abcdefghij",
    "",
    "2.5",
    "An already registered login, surrounded by leading/trailing blanks, was accepted as login for a new member" );
nbTests++;
nbErrors += addMemberAlreadyExistsTest (
    sn,
    "An" + "toi" + "ne",
    "abcdefghij",
    "",
    "2.6",
    "A String concatenation building an already registered login was accepted as login for a new member" );

nbTests++;
// check that 'sn' was not modified
if (nbFilms != sn.nbfilms()) {
    System.out
        .println("Error : the number of films was unexpectedly changed by addMember()");
    nbErrors++;
}
nbTests++;
if (nbBooks != sn.nbBooks()) {
    System.out
        .println("Error : the number of books was unexpectedly changed by addMember()");
    nbErrors++;
}

// Display final state of 'sn'
System.out.println("Final state of the social network : " + sn);

// Print a summary of the tests and return test results
try{
    TestReport tr = new TestReport(nbTests, nbErrors);
    System.out.println("AddMemberTest:" + tr);
    return tr;
}
catch (NotTestReportException e){ //This shouldn't happen
    System.out.println("Unexpected error in AddMemberTest test code – Can't return valuable test results");
    return null;
}
}

/**
 * Launches test()
 * @param args not used
 */
public static void main(String[] args) {
    test();
}
}

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